Supplementary Document

MTDriverGNN: Multitask Learning Graph Neural Network for Cancer Driver Gene Prioritization

Table 1. Comparison of MTDriverGNN with baseline methods across 12 cancer types, evaluated using AUPRC (mean ± standard deviation).

Cancer	MTDriverGNN	EMOGI	MTGCN	DGHNN
BRCA	0.7306 ± 0.0096	0.6482 ± 0.0045	0.6583 ± 0.0040	0.6832 ± 0.0062
BLCA	0.7155 ± 0.0219	0.5485 ± 0.0101	0.6568 ± 0.0077	0.6935 ± 0.0064
LUAD	0.6653 ± 0.0096	0.5591 ± 0.0105	0.6279 ± 0.0099	0.6410 ± 0.0091
LIHC	0.5076 ± 0.0196	0.3845 ± 0.0193	0.4645 ± 0.0195	0.4811 ± 0.0191
THCA	0.3533 ± 0.0153	0.2587 ± 0.0084	0.2907 ± 0.0110	0.3145 ± 0.0233
LUSC	0.4461 ± 0.0536	0.2222 ± 0.0266	0.3099 ± 0.0407	0.4054 ± 0.0542
ESCA	0.5027 ± 0.0289	0.4174 ± 0.0297	0.4721 ± 0.0279	0.4763 ± 0.0145
PRAD	0.7468 ± 0.0256	0.5233 ± 0.0220	0.6143 ± 0.0189	0.7058 ± 0.0129
STAD	0.7240 ± 0.0168	0.4604 ± 0.0153	0.5931 ± 0.0117	0.6449 ± 0.0083
COAD	0.5353 ± 0.0205	0.3459 ± 0.0067	0.3816 ± 0.0085	0.4468 ± 0.0220
UCEC	0.7121 ± 0.0235	0.4020 ± 0.0143	0.4954 ± 0.0145	0.5342 ± 0.0250
CESC	0.5715 ± 0.0687	0.5547 ± 0.0522	0.5972 ± 0.0528	0.5635 ± 0.0414

Table 2. Performance of Single-task and Multi-task settings across 12 cancer types, evaluated using AUPRC (mean ± standard deviation).

Cancer	Single-task	Multi-task
BRCA	0.6233 ± 0.0100	0.7110 ± 0.0125
BLCA	0.7104 ± 0.0114	0.7028 ± 0.0191
LUAD	0.6473 ± 0.0052	0.6508 ± 0.0157
LIHC	0.4556 ± 0.0209	0.4863 ± 0.0226
THCA	0.2435 ± 0.0144	0.3107 ± 0.0309
LUSC	0.3054 ± 0.0377	0.4197 ± 0.0402
ESCA	0.4000 ± 0.0429	0.4926 ± 0.0186
PRAD	0.6577 ± 0.0390	0.7021 ± 0.0315
STAD	0.6796 ± 0.0261	0.6938 ± 0.0235
COAD	0.3978 ± 0.0237	0.5136 ± 0.0212
UCEC	0.5750 ± 0.0190	0.6837 ± 0.0238
CESC	0.5098 ± 0.0521	0.4628 ± 0.0281

Table 3. Comparison of MTDriverGNN without and with pretraining across 12 cancer types, evaluated using AUPRC (mean ± standard deviation).

Cancer	W/O Pretraining	Pretraining
BRCA	0.7110 ± 0.0125	0.7306 ± 0.0096
BLCA	0.7028 ± 0.0191	0.7155 ± 0.0219
LUAD	0.6508 ± 0.0157	0.6653 ± 0.0096
LIHC	0.4863 ± 0.0226	0.5076 ± 0.0196
THCA	0.3107 ± 0.0309	0.3533 ± 0.0153
LUSC	0.4197 ± 0.0402	0.4461 ± 0.0536
ESCA	0.4926 ± 0.0186	0.5027 ± 0.0289
PRAD	0.7021 ± 0.0315	0.7468 ± 0.0256
STAD	0.6938 ± 0.0235	0.7240 ± 0.0168
COAD	0.5136 ± 0.0212	0.5353 ± 0.0205
UCEC	0.6837 ± 0.0238	0.7121 ± 0.0235
CESC	0.4628 ± 0.0281	0.5715 ± 0.0687

Table 4. Summary statistics of feature counts and gene counts across 12 cancer types.

	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
Mutation-derived features	4	4	4	4	4	4	4	4	4	4	4	4
Pathway-derived features	8	6	7	5	11	7	5	8	8	9	8	14
Total features	12	10	11	9	15	11	9	12	12	13	12	18
Driver genes	202	95	179	64	82	26	71	50	73	141	68	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187